

Online Appendix 1: Technical Proofs for Covariance-Scale Calibration of Empirical Populations

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Editor:

Abstract

This Online Appendix contains the complete proofs of the uniqueness, minimum-displacement, first-order projection-equivalence, and conditional bootstrap-premise results stated in the accompanying manuscript. The main paper retains theorem statements, proof sketches, and the principal arguments; the derivations collected here provide the formal proofs.

Keywords: covariance-scale calibration; technical proofs

1 Proofs of Geometric and Minimum-Distortion Properties

This appendix provides the formal proofs for the uniqueness and minimal distortion optimality of the Surrogate Population framework introduced in Section 2.

1.1 Proof of Sample-Conditional Affine Uniqueness

Proof This proof establishes the sample-conditional coordinate-wise affine uniqueness result stated as Theorem 2 in the accompanying main manuscript.

Let the calibrated observations be $\mathbf{y}_i = \mathbf{A}\mathbf{x}_i + \mathbf{b}$. First, apply Condition 1 (Mean preservation). The empirical mean of the transformed data is:

$$\bar{\mathbf{y}} = \frac{1}{n} \sum_{i=1}^n (\mathbf{A}\mathbf{x}_i + \mathbf{b}) = \mathbf{A}\bar{\mathbf{x}} + \mathbf{b} \quad (1)$$

Setting $\bar{\mathbf{y}} = \bar{\mathbf{x}}$ yields $\mathbf{b} = (\mathbf{I}_p - \mathbf{A})\bar{\mathbf{x}}$. Consequently, the centered observations simplify to:

$$\mathbf{y}_i - \bar{\mathbf{y}} = \mathbf{A}(\mathbf{x}_i - \bar{\mathbf{x}}) \quad (2)$$

Second, apply Condition 3 (Order preservation). For f to preserve the coordinate-wise ordering of the data (i.e., for any feature j , $x_{ij} \leq x_{kj} \implies y_{ij} \leq y_{kj}$ without interference from other features), the transformation matrix $\mathbf{A}$ must be a positive diagonal matrix. Let $\mathbf{A} = \text{diag}(a_1, a_2, \dots, a_p)$ where $a_j > 0$ for all $j \in \{1, \dots, p\}$.

The positive-marginal-variance condition is essential for identification: when $(\mathbf{S}_{\text{emp}})_{jj} = 0$, the covariance equation imposes no restriction on a_j . The following argument therefore applies coordinatewise on the non-degenerate coordinates for which $(\mathbf{S}_{\text{emp}})_{jj} > 0$.

Third, apply Condition 2 (Covariance calibration). The empirical covariance of $\mathbf{Y}$ is:

$$\mathbf{S}_Y = \frac{1}{n} \sum_{i=1}^n \mathbf{A}(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{A}^\top = \mathbf{A} \mathbf{S}_{\text{emp}} \mathbf{A}^\top \quad (3)$$

We require $\mathbf{S}_Y = \mathbf{S} = \frac{n}{n-1} \mathbf{S}_{\text{emp}}$. Thus, we must have:

$$\mathbf{A} \mathbf{S}_{\text{emp}} \mathbf{A}^\top = \frac{n}{n-1} \mathbf{S}_{\text{emp}}. \quad (4)$$

Because $\mathbf{A}$ is diagonal, the (j, k) -th element of the resulting matrix equation is:

$$a_j a_k (\mathbf{S}_{\text{emp}})_{jk} = \frac{n}{n-1} (\mathbf{S}_{\text{emp}})_{jk} \quad (5)$$

Taking $j = k$ in the covariance identity for the observed sample gives

$$a_j^2 (\mathbf{S}_{\text{emp}})_{jj} = \frac{n}{n-1} (\mathbf{S}_{\text{emp}})_{jj}.$$

Because $(\mathbf{S}_{\text{emp}})_{jj} > 0$, it follows that

$$a_j^2 = \frac{n}{n-1} \implies a_j = \sqrt{\frac{n}{n-1}} = s$$

Since $a_j > 0$ by Condition 3, the unique solution for the transformation matrix is the isotropic scalar matrix $\mathbf{A} = s \mathbf{I}_p$. Substituting $\mathbf{A}$ back into the translation vector yields $\mathbf{b} = (1 - s) \bar{\mathbf{x}}$. The unique transformation is therefore $\mathbf{y}_i = s \mathbf{x}_i + (1 - s) \bar{\mathbf{x}} = s(\mathbf{x}_i - \bar{\mathbf{x}}) + \bar{\mathbf{x}}$, which completes the proof. $\blacksquare$

1.2 Proof of Fixed-Correspondence Minimum Distortion

Proof This proof establishes the fixed-correspondence minimum-distortion result stated as Theorem 4 in the accompanying main manuscript. Let

$$\mathbf{X}_c = \begin{bmatrix} (\mathbf{x}_1 - \bar{\mathbf{x}})^\top \\ \vdots \\ (\mathbf{x}_n - \bar{\mathbf{x}})^\top \end{bmatrix}, \quad \mathbf{Y}_c = \begin{bmatrix} (\mathbf{y}_1 - \bar{\mathbf{x}})^\top \\ \vdots \\ (\mathbf{y}_n - \bar{\mathbf{x}})^\top \end{bmatrix}.$$

The mean constraint implies that both matrices are centered. The covariance constraint is equivalent to

$$\frac{1}{n} \mathbf{Y}_c^\top \mathbf{Y}_c = \mathbf{S} = s^2 \mathbf{S}_{\text{emp}} = \frac{s^2}{n} \mathbf{X}_c^\top \mathbf{X}_c, \quad s = \sqrt{\frac{n}{n-1}},$$

and hence

$$\mathbf{Y}_c^\top \mathbf{Y}_c = s^2 \mathbf{X}_c^\top \mathbf{X}_c.$$

First, suppose that $\mathbf{S}_{\text{emp}} = \mathbf{0}$. Since $\mathbf{S}_{\text{emp}}$ is positive semidefinite, this implies $\mathbf{x}_i = \bar{\mathbf{x}}$ for all i . The target covariance $\mathbf{S} = s^2 \mathbf{S}_{\text{emp}}$ is also zero, forcing $\mathbf{y}_i = \bar{\mathbf{x}}$ for all i . Thus, $\mathbf{Y} = \mathbf{X}$ is the unique feasible point configuration.

Now suppose that $\mathbf{S}_{\text{emp}} \neq \mathbf{0}$. The covariance constraint fixes the Frobenius norm of every feasible candidate matrix:

$$\|\mathbf{Y}_c\|_F = s\|\mathbf{X}_c\|_F, \quad s = \sqrt{\frac{n}{n-1}}.$$

The total squared pointwise distortion is

$$\sum_{i=1}^n \|\mathbf{y}_i - \mathbf{x}_i\|_2^2 = \|\mathbf{Y}_c - \mathbf{X}_c\|_F^2 = \|\mathbf{Y}_c\|_F^2 + \|\mathbf{X}_c\|_F^2 - 2\text{tr}(\mathbf{Y}_c^\top \mathbf{X}_c).$$

Since the first two norm terms are constant over all feasible candidate configurations, minimizing total Euclidean distortion is equivalent to maximizing the Frobenius inner product

$$\langle \mathbf{Y}_c, \mathbf{X}_c \rangle_F = \text{tr}(\mathbf{Y}_c^\top \mathbf{X}_c).$$

By the Frobenius Cauchy–Schwarz inequality,

$$\text{tr}(\mathbf{Y}_c^\top \mathbf{X}_c) \leq \|\mathbf{Y}_c\|_F \|\mathbf{X}_c\|_F = s\|\mathbf{X}_c\|_F^2.$$

The upper bound is attained by

$$\mathbf{Y}_c = s\mathbf{X}_c,$$

which satisfies the covariance constraint.

To establish uniqueness, equality in the Cauchy–Schwarz bound holds if and only if

$$\mathbf{Y}_c = c\mathbf{X}_c$$

for some scalar c . The covariance constraint requires $c^2 = s^2$, whereas maximizing

$$\text{tr}(\mathbf{Y}_c^\top \mathbf{X}_c) = c\|\mathbf{X}_c\|_F^2$$

uniquely selects the positive root

$$c = s = \sqrt{\frac{n}{n-1}} > 0.$$

Hence,

$$\mathbf{Y}_c = s\mathbf{X}_c$$

is the unique optimal centered configuration. Re-adding the sample mean $\bar{\mathbf{x}}$ yields

$$\mathbf{y}_i = \bar{\mathbf{x}} + s(\mathbf{x}_i - \bar{\mathbf{x}}), \quad i = 1, \dots, n,$$

which proves that the surrogate population is the unique minimizer of the stated fixed-correspondence pointwise squared-Euclidean distortion problem. $\blacksquare$

2 Proofs of Projection Equivalence and Conditional Bootstrap Transfer

This appendix provides the formal proofs for first-order projection equivalence and for preservation of the measure-level premises in the assumed conditional bootstrap-validity statement.

2.1 Proof of First-Order Projection Equivalence

Proof This proof establishes the projection-equivalence result stated as Theorem 10 in the accompanying main manuscript.

Recall that the surrogate transformation is $\mathbf{y}_i = s\mathbf{x}_i + (1-s)\bar{\mathbf{x}}$, where the scaling factor is $s = \sqrt{\frac{n}{n-1}}$. The residual perturbation for each observation is:

$$\Delta_i = \mathbf{y}_i - \mathbf{x}_i = (s-1)(\mathbf{x}_i - \bar{\mathbf{x}})$$

By the binomial expansion of $(1 - \frac{1}{n})^{-1/2}$, we have $s = 1 + \frac{1}{2n} + \mathcal{O}(n^{-2})$, which implies $s-1 = \frac{1}{2n} + \mathcal{O}(n^{-2})$.

The Hájek projection of the statistic evaluated on the transformed data points is given by $\hat{L}_n(\mathbf{Y}) = \frac{m}{n} \sum_{i=1}^n g_P(\mathbf{y}_i)$. Because the fixed population projection g_P has a globally Lipschitz gradient, a second-order Taylor expansion around $\mathbf{x}_i$ gives

$$g_P(\mathbf{y}_i) = g_P(\mathbf{x}_i) + \nabla g_P(\mathbf{x}_i)^\top \Delta_i + R_i, \quad |R_i| \leq \frac{L}{2} \|\Delta_i\|_2^2,$$

where L is a global Lipschitz constant for ∇g_P . Summing over all observations and scaling by $\sqrt{n}$, the difference between the projections is:

$$\begin{aligned} \sqrt{n} \left(\hat{L}_n(\mathbf{Y}) - \hat{L}_n(\mathbf{X}) \right) &= \sqrt{n} \frac{m}{n} \sum_{i=1}^n \left(\nabla g_P(\mathbf{x}_i)^\top \Delta_i + \mathcal{R}_i \right) \\ &= m\sqrt{n}(s-1) \left[\frac{1}{n} \sum_{i=1}^n \nabla g_P(\mathbf{x}_i)^\top (\mathbf{x}_i - \bar{\mathbf{x}}) \right] + m\sqrt{n} \left[\frac{1}{n} \sum_{i=1}^n \mathcal{R}_i \right] \end{aligned}$$

We now analyze the two bracketed terms in probability. First, consider the linear gradient term. Write

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n \nabla g_P(\mathbf{x}_i)^\top (\mathbf{x}_i - \bar{\mathbf{x}}) &= \frac{1}{n} \sum_{i=1}^n \nabla g_P(\mathbf{x}_i)^\top (\mathbf{x}_i - \boldsymbol{\mu}) \\ &\quad - (\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \left\{ \frac{1}{n} \sum_{i=1}^n \nabla g_P(\mathbf{x}_i) \right\}. \end{aligned}$$

By the Cauchy–Schwarz inequality,

$$\mathbb{E}_P \left| \nabla g_P(\mathbf{X})^\top (\mathbf{X} - \boldsymbol{\mu}) \right| \leq \left\{ \mathbb{E}_P \|\nabla g_P(\mathbf{X})\|_2^2 \right\}^{1/2} \left\{ \mathbb{E}_P \|\mathbf{X} - \boldsymbol{\mu}\|_2^2 \right\}^{1/2} < \infty.$$

Hence, the weak law of large numbers gives

$$\frac{1}{n} \sum_{i=1}^n \nabla g_P(\mathbf{x}_i)^\top (\mathbf{x}_i - \boldsymbol{\mu}) \xrightarrow{P} \mathbb{E}_P \left[\nabla g_P(\mathbf{X})^\top (\mathbf{X} - \boldsymbol{\mu}) \right].$$

Moreover,

$$\bar{\mathbf{x}} - \boldsymbol{\mu} = o_P(1),$$

and

$$\frac{1}{n} \sum_{i=1}^n \nabla g_P(\mathbf{x}_i) = O_P(1),$$

because

$$\mathbb{E}_P \|\nabla g_P(\mathbf{X})\|_2 < \infty,$$

which follows from $\mathbb{E}_P \|\nabla g_P(\mathbf{X})\|_2^2 < \infty$. Therefore,

$$(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \left\{ \frac{1}{n} \sum_{i=1}^n \nabla g_P(\mathbf{x}_i) \right\} = o_P(1).$$

Consequently,

$$\frac{1}{n} \sum_{i=1}^n \nabla g_P(\mathbf{x}_i)^\top (\mathbf{x}_i - \bar{\mathbf{x}}) \xrightarrow{P} \mathbb{E}_P \left[\nabla g_P(\mathbf{X})^\top (\mathbf{X} - \boldsymbol{\mu}) \right] \equiv C,$$

where C is finite. Since

$$m\sqrt{n}(s-1) = m\sqrt{n} \left\{ \frac{1}{2n} + O(n^{-2}) \right\} = O(n^{-1/2}),$$

the linear-gradient term is $o_P(1)$.

Second, consider the remainder term. We bound the average of the absolute remainders:

$$\frac{1}{n} \sum_{i=1}^n |\mathcal{R}_i| \leq \frac{L}{2} (s-1)^2 \frac{1}{n} \sum_{i=1}^n \|\mathbf{x}_i - \bar{\mathbf{x}}\|^2 = \frac{L}{2} \mathcal{O}(n^{-2}) \text{tr}(\mathbf{S}_{\text{emp}}) = \mathcal{O}_P(n^{-2})$$

Scaling this remainder average by $m\sqrt{n}$ yields $\mathcal{O}_P(n^{-3/2})$, which trivially vanishes as $n \rightarrow \infty$. Both terms converge to zero, concluding the proof that $\sqrt{n}(\hat{L}_n(\mathbf{Y}) - \hat{L}_n(\mathbf{X})) = o_P(1)$. ■

2.2 Proof of Conditional Bootstrap-Premise Preservation

Proof This proof establishes the conditional bootstrap-premise preservation result stated as Theorem 11 in the accompanying main manuscript.

Let

$$\hat{P}_n^X = \frac{1}{n} \sum_{i=1}^n \delta_{\mathbf{x}_i}, \quad \hat{P}_n^Y = \frac{1}{n} \sum_{i=1}^n \delta_{\mathbf{y}_i}$$

denote the empirical measures associated with the original and surrogate samples, respectively. We verify that $\hat{P}_n^Y$ satisfies the two measure-level conditions in the conditional bootstrap-stability assumption of Theorem 11 in the accompanying main manuscript.

Throughout this proof, g_P denotes the population first-order projection associated with the fixed distribution P :

$$g_P(\mathbf{x}) = \mathbb{E}_P[h(\mathbf{x}, \mathbf{X}_2, \dots, \mathbf{X}_m)] - \theta(P).$$

In particular, g_P is not redefined after the original empirical measure $\hat{P}_n^X$ is replaced by the surrogate empirical measure $\hat{P}_n^Y$.

First, couple $\hat{P}_n^X$ and $\hat{P}_n^Y$ by pairing $\mathbf{x}_i$ with $\mathbf{y}_i$ for each $i = 1, \dots, n$. This coupling gives

$$W_2^2(\hat{P}_n^X, \hat{P}_n^Y) \leq \frac{1}{n} \sum_{i=1}^n \|\mathbf{y}_i - \mathbf{x}_i\|_2^2.$$

By the average-perturbation bound established in the accompanying main manuscript,

$$\frac{1}{n} \sum_{i=1}^n \|\mathbf{y}_i - \mathbf{x}_i\|_2^2 = \mathcal{O}_P(n^{-2}),$$

and therefore

$$W_2(\hat{P}_n^X, \hat{P}_n^Y) = \mathcal{O}_P(n^{-1}).$$

Under the assumed finite-second-moment condition,

$$\mathbb{E}_P \|\mathbf{X}\|_2^2 < \infty,$$

and for fixed ambient dimension p , the empirical measure $\hat{P}_n^X$ converges to P in the quadratic Wasserstein metric:

$$W_2(\hat{P}_n^X, P) \longrightarrow 0 \quad \text{almost surely,}$$

and therefore also in probability. Equivalently, in $\mathcal{P}_2(\mathbb{R}^p)$, weak convergence together with convergence of second moments yields W_2 -convergence; see Villani (2009). Hence, by the triangle inequality,

$$\begin{aligned} W_2(\hat{P}_n^Y, P) &\leq W_2(\hat{P}_n^Y, \hat{P}_n^X) + W_2(\hat{P}_n^X, P) \\ &\xrightarrow{P} 0. \end{aligned}$$

Consequently,

$$\hat{P}_n^Y \Rightarrow P \quad \text{in probability.}$$

Next, we verify convergence of the empirical second moment of the fixed population projection score g_P . Let

$$\Delta_i = \mathbf{y}_i - \mathbf{x}_i = (s - 1)(\mathbf{x}_i - \bar{\mathbf{x}}).$$

A Taylor expansion of g_P around $\mathbf{x}_i$ gives

$$g_P(\mathbf{y}_i) - g_P(\mathbf{x}_i) = \nabla g_P(\mathbf{x}_i)^\top \Delta_i + R_i, \quad |R_i| \leq \frac{L}{2} \|\Delta_i\|_2^2,$$

where L is a global Lipschitz constant for ∇g_P . Therefore,

$$\{g_P(\mathbf{y}_i) - g_P(\mathbf{x}_i)\}^2 \leq 2\|\nabla g_P(\mathbf{x}_i)\|_2^2 \|\Delta_i\|_2^2 + \frac{L^2}{2} \|\Delta_i\|_2^4.$$

Because

$$s - 1 = \mathcal{O}(n^{-1}),$$

the fourth-order perturbation average satisfies

$$\frac{1}{n} \sum_{i=1}^n \|\Delta_i\|_2^4 = (s-1)^4 \frac{1}{n} \sum_{i=1}^n \|\mathbf{x}_i - \bar{\mathbf{x}}\|_2^4 = \mathcal{O}_P(n^{-4}),$$

where the final order follows from

$$\mathbb{E}_P \|\mathbf{X}\|_2^4 < \infty.$$

Moreover, by the Cauchy–Schwarz inequality,

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n \|\nabla g_P(\mathbf{x}_i)\|_2^2 \|\Delta_i\|_2^2 &\leq \left\{ \frac{1}{n} \sum_{i=1}^n \|\nabla g_P(\mathbf{x}_i)\|_2^4 \right\}^{1/2} \left\{ \frac{1}{n} \sum_{i=1}^n \|\Delta_i\|_2^4 \right\}^{1/2} \\ &= \mathcal{O}_P(n^{-2}), \end{aligned}$$

where the final equality uses

$$\mathbb{E}_P \|\nabla g_P(\mathbf{X})\|_2^4 < \infty.$$

It follows that

$$\frac{1}{n} \sum_{i=1}^n \{g_P(\mathbf{y}_i) - g_P(\mathbf{x}_i)\}^2 = \mathcal{O}_P(n^{-2}).$$

Consequently,

$$\begin{aligned} \left| \frac{1}{n} \sum_{i=1}^n g_P^2(\mathbf{y}_i) - \frac{1}{n} \sum_{i=1}^n g_P^2(\mathbf{x}_i) \right| &\leq \left\{ \frac{1}{n} \sum_{i=1}^n [g_P(\mathbf{y}_i) - g_P(\mathbf{x}_i)]^2 \right\}^{1/2} \\ &\quad \times \left\{ \frac{1}{n} \sum_{i=1}^n [g_P(\mathbf{y}_i) + g_P(\mathbf{x}_i)]^2 \right\}^{1/2}. \end{aligned}$$

The first factor is $\mathcal{O}_P(n^{-1})$. For the second factor,

$$\frac{1}{n} \sum_{i=1}^n \{g_P(\mathbf{y}_i) + g_P(\mathbf{x}_i)\}^2 \leq 2 \frac{1}{n} \sum_{i=1}^n g_P^2(\mathbf{y}_i) + 2 \frac{1}{n} \sum_{i=1}^n g_P^2(\mathbf{x}_i).$$

The second term is $\mathcal{O}_P(1)$ by the assumed finite second moment of $g_P(\mathbf{X})$. Moreover, by

$$g_P^2(\mathbf{y}_i) \leq 2\{g_P(\mathbf{y}_i) - g_P(\mathbf{x}_i)\}^2 + 2g_P^2(\mathbf{x}_i),$$

the preceding average-squared perturbation bound and the same moment condition give

$$\frac{1}{n} \sum_{i=1}^n g_P^2(\mathbf{y}_i) = \mathcal{O}_P(1).$$

Hence the second factor is $\mathcal{O}_P(1)$, and therefore

$$\left| \frac{1}{n} \sum_{i=1}^n g_P^2(\mathbf{y}_i) - \frac{1}{n} \sum_{i=1}^n g_P^2(\mathbf{x}_i) \right| = o_P(1).$$

Since

$$\frac{1}{n} \sum_{i=1}^n g_P^2(\mathbf{x}_i) \xrightarrow{P} \mathbb{E}_P[g_P^2(\mathbf{X})],$$

we obtain

$$\int g_P^2 d\widehat{P}_n^Y = \frac{1}{n} \sum_{i=1}^n g_P^2(\mathbf{y}_i) \xrightarrow{P} \mathbb{E}_P[g_P^2(\mathbf{X})].$$

We have therefore verified the two measure-level premises required by the assumed conditional bootstrap-stability statement:

$$\widehat{P}_n^Y \Rightarrow P \quad \text{in probability,}$$

and

$$\int g_P^2 d\widehat{P}_n^Y \xrightarrow{P} \mathbb{E}_P[g_P^2(\mathbf{X})].$$

Consequently, whenever the stated statistic-specific conditional bootstrap-validity theorem applies to empirical-measure sequences satisfying these two premises, it applies with

$$Q_n = \widehat{P}_n^Y.$$

The surrogate construction does not independently establish bootstrap consistency for arbitrary statistics or learning procedures. ■

References

C. Villani. *Optimal Transport: Old and New*, volume 338 of *Grundlehren der mathematischen Wissenschaften*. Springer-Verlag, Berlin, 2009. ISBN 978-3-540-71049-3.